Page-12 FIRST PRE-BOARD EXAM. (2024-2025)
CLASS: XII
SUBJECT: INFORMATICS PRACTICES (065)

Time allowed: 3 Hours **Maximum Marks: 70**

General Instructions :

- This question paper contains 37 questions.
- All questions are compulsory. However, internal choices have been provided in some questions. Attempt only one of the choices in such questions
- The paper is divided into 5 Sections- A, B, C, D and E.
- Section A consists of 21 questions (1 to 21). Each question carries 1 Mark.
- Section B consists of 7 questions (22 to 28). Each question carries 2 Marks.
- Section C consists of 4 questions (29 to 32). Each question carries 3 Marks.
- Section D consists of 2 questions (33 to 34). Each question carries 4 Marks.
- Section E consists of 3 questions (35 to 37). Each question carries 5 Marks.
- All programming questions are to be answered using Python Language only.
- In case of MCQ, text of the correct answer should also be written.

	Section-A (21 x 1 = 21 Marks)	
1	In pandas, a Series can only hold data of one data type.	1
2	Which of the following statements correctly uses the WHERE clause? A) SELECT * FROM Employees WHERE Salary > 50000; B) SELECT * FROM Employees WHERE > 50000; C) SELECT * FROM Employees WHERE Salary 50000; D) SELECT * FROM Employees WHERE Salary =;	1
3	What is the primary function of a network repeater? A) To filter data packets B) To amplify and regenerate signals C) To provide network security D) To route data between different networks	1
4	Which of the following SQL statements correctly deletes all records from a table named Customers? A) DELETE Customers;	1

	B) DELETE FROM Customers; C) REMOVE FROM Customers; D) DROP Customers;	
5	Which of the following is a recommended method for disposing of e-waste? A) Throwing it in the regular trash B) Recycling through certified e-waste recyclers C) Burning it in open spaces D) Burying it in landfills	1
6	Which of the following methods is used to select a single column from a pandas DataFrame named df? A) df.select('column_name') B) df['column_name'] C) df.column_name D) Both B and C	1
7	Which library is primarily used for plotting graphs in Python? A) NumPy B) Matplotlib C) Pandas D) Seaborn	1
8	The HAVING clause can only be used with aggregate functions like SUM(), COUNT(), and AVG().	1
9	Which function is used to read a CSV file into a pandas DataFrame? A) pd.load_csv() B) pd.read_csv() C) pd.import_csv() D) pd.open_csv()	1
10	Which of the following is NOT a form of plagiarism? A) Copying text word-for-word without citation B) Paraphrasing someone else's ideas without attribution C) Citing sources correctly in a bibliography D) Submitting someone else's work as your own	1
11	The COUNT() function can be used with a _____ clause to count the number of rows that meet specific criteria. A) GROUP BY B) WHERE C) ORDER BY D) HAVING	1
12	In a bus topology, what happens if the main cable (the bus) fails? A) Only the device at the end fails B) The entire network goes down C) Data transmission speed increases D) Other devices continue to function normally	1
13	What is the default number of rows returned by df.tail() if no argument is provided? A) 1 B) 5 C) 10 D) 0	1

14	What is phishing? A) A method of fishing in deep waters B) A cyber attack where attackers impersonate legitimate organizations to steal sensitive information C) A type of software used to enhance internet security D) A technique for improving email marketing	1
15	Which of the following statements is true about creating a Series from a dictionary? A) The keys of the dictionary become the index of the Series. B) The values of the dictionary become the index of the Series. C) You cannot create a Series from a dictionary. D) The Series will always be ordered by the keys.	1
16	Match the following SQL functions/clauses with their descriptions :	1

SQL function		Description	
P	ROUND()	1	Returns a string with all uppercase letters
Q	UPPER()	2	Returns the highest value in a column
R	MAX()	3	Returns the current date and time
S	NOW()	4	Returns a rounded numeric value

A) P-1,Q-2,R-3,S-4
B) P-4,Q-3,R-2,S-1
C) P-4,Q-1,R-2,S-3
D) P-2,Q-1,R-4,S-3

17	When using boolean indexing, you can combine multiple conditions using _____ operators. A) logical B) arithmetic C) comparison D) bitwise	1
18	What is the default orientation of a bar plot created using plt.bar()? A) Horizontal B) Vertical C) Diagonal D) Stacked	1
19	Which type of network is most likely to use Bluetooth technology? A) LAN B) WAN C) MAN D) PAN	1

	Q20 and Q21 are Assertion(A) and Reason(R) based questions. Mark the correct choice as: (A) Both A and R are true and R is the correct explanation for A (B) Both A and R are true and R is not the correct explanation for A (C) A is True but R is False (D) A is False but R is True	
20	Assertion (A): The loc[] method in pandas is used for label-based indexing. Reason (R): The iloc[] method is used for positional indexing.	1
21	Assertion (A) : DDL commands are used to define and modify database structures. Reason (R): DML commands are used to query and manipulate data within those structures.	

	Section-B (7 x 2=14 Marks)	
22	(A) What is the primary purpose of the Pandas library in Python, and how does it facilitate data manipulation? OR (B) Explain the difference between a Pandas Series and a DataFrame.	2
23	What do you understand by plagiarism? Mention any two ways to avoid plagiarism.	2
24	Consider the given SQL string: “12#All the Best!” Write suitable SQL queries for the following: i. Returns the position of the first occurrence of the substring “the” in the given string. ii. To extract last five characters from the string.	2
25	(A) What is a website? What is the difference between a website and a webpage? OR (B) What is a web browser? What is the purpose of add-ons in a web browser?	2
26	Define the term Candidate Key in a database. Explain how it is different from Alternate Key.	2
27	Mention any two health hazards associated with inappropriate and excessive use of gadgets.	2

28	(A) Abhay is writing a Python program to create a Dataframe. The python code written by him has errors in it. Rewrite the correct code and underline the corrections made. Import pandas as pd df ={"Technology":["Programming","Robotics","3D Printing"],"Time(in months)":[4,4,3]} df= Pd.dataframe(df) Print(df) OR (B) Complete the given Python code to get the required output as: **Rajasthan** import _____ as pd di = {'Corbett': 'Uttarakhand', 'Sariska': 'Rajasthan', 'Kanha': 'Madhya Pradesh’, 'Gir':'Gujarat'} NP = _____. Series(_____) print(NP[_____])	2

	Section-C (3 x 4 = 12 Marks)	
29	Infotech Cyber Café is replacing their old computers with a new one. The Café decided to dump the old computer in a nearby empty field/plot. I. Explain any one potential environmental hazard associated with improper e-waste disposal. II. Suggest one responsible way to Infotech Cyber Café for proper disposal of their old computers. III. Describe the importance of recycling in e-waste management.	3
30	Write a Python program to create the following DataFrame using a list of dictionaries.	3

	Course	Fee
0	MCA	50000
1	BCA	40000
2	PGDCA	20000
3	MSc	35000

OR

Write a Python Program to create a Pandas Series as shown below using a dictionary. Note that the left column indicates the indices and the right column displays the data.

Gujarat	Gandhinagar
Rajasthan	Jaipur
Assam	Dispur
UP	Lucknow

31 I. Write an SQL statement to create a table named CRICKET, with the following specifications: **3**

Column	Data Type	Key
Id	Numeric	Primary Key
Name	Varchar(15)	
Position	Varchar(15)	
Debue	Date	
Avg	Float(10,2)	

II. Write SQL Query to insert the following data in the CRIKCET Table
(101, Virat Kohli, Opener,2010-10-12,55.5)

32 Consider the following tables: **3**
Table 1:
STUDENT which stores Student ID (S_ID), Student Name (S_NAME), Student House (S_House)
Table 2:
RESULT which stores Student ID (S_ID), Class (S_Class), Marks (S_Marks) and Result (S_Result) for various Students.
Note: Attribute names are written within brackets.
Table: STUDENT

S_ID	S_Name	S_House
101	Sanjay Sharma	Raman
102	Mukesh Rathi	Tagore
103	Suman Verma	Shivaji
104	Akhilesh	Ashoka
105	Anju	Raman

Table: RESULT

S_ID	S_Class	S_Marks	S_Result
101	12	480	Pass
102	11	320	Pass
103	11	100	Fail
104	10	250	Pass
105	10	300	Pass

Write appropriate SQL queries for the following:
I. Display Total Number of students appeared class-wise.
II. Display name of all students in UPPER Case who belongs to Raman house.
III. Display Name of Students along with their corresponding Class & result.

OR

Consider the following tables:
Table 1:
TEACHER, which stores T_ID, T_Name, T_State. The table displays basic information of the TEACHER
Table 2:
SCHOOL, which stores T_ID, T_Post, T_Subject and T_Basic. The table displays the Basic Salary of each teacher.

Table: TEACHER

T_ID	T_Name	T_State
11	Narendra Kumar	Delhi
12	Manoj Singh	MP
13	Amit Pandey	UP
14	Nisha Verma	Bihar
15	Sachin Sharma	HP

Table: SCHOOL

T_ID	T_Post	T_Subject	T_Basic
11	PGT	Hindi	55000
12	PGT	IP	60000
13	TGT	Bio	50000
14	TGT	English	45000
15	PRT	Music	40000

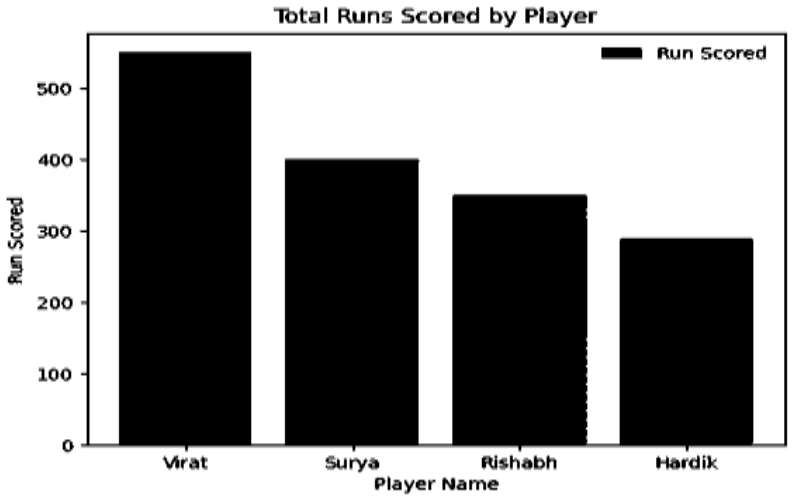
I. Display the post-wise average basic salary.
II. Display the all subjects in increasing order of basic salary.
III. Display all the name of teachers along with their post & basic salary.

Section-D (4 x 2 = 8 Marks)

33 During the IPL season, an Analyst has to fill in the blanks in a Python program that generates a bar chart. This bar chart represents the Total Runs Scored by Player. 4

Player Name	Run Scored
Virat	550
Surya	400
Rishabh	350
Hardik	290

Help analyst to complete the code.




```
import _____________ as plt #Statement-1
Players = ['Virat', 'Surya', 'Rishabh', 'Hardik']
Run_Scored = [550,400,350, 290]
plt.bar( Players, __________, label='Run Scored') #Statement-2
plt._________('Player Name') #Statement-3
plt.ylabel('Run Scored')
plt.legend()
plt.title(____________________) #Statement-4
plt.show()
```

I. Write the suitable code for the import statement in the blank space in the line marked as Statement-1.

II. Refer to the graph shown above and fill in the blank in Statement-2 with suitable Python code.

III. Fill in the blank in Statement-3 with the name of the function to set the label on the x-axis.

IV. Refer the graph shown above and fill the blank in Statement-4 with suitable Chart Title.

34	(A) Rajeev, who works as a database designer, has developed a database for a Grocery Store. This database	4

includes a table ‘Billing’ whose column (attribute) names are mentioned below:

Item_ID: Shows the unique code for each item.
Item_Name: Indicates the item’s name.
Item_Brand: Specifies the item brand name.
Item_Price: Specifies the item price.

Table: Billing

Item_ID	Item_Name	Item_Brand	Item_Price
P001	Liquid Soap	Dettol	200
P002	Cold Drink	Pepsi	40
P003	Battery	Duracell	50
P004	Deodorant	Nivia	180
P005	Shampoo	Loreal	330

Write the output for following queries:

I. SELECT LOWER(Item_Name) FROM Billling;
II. SELECT MAX(Item_Price) FROM Billing;
III. SELECT LENGTH(Item_Brand) FROM Billing;
IV. SELECT Item_ID, Item_Price FROM Billing ORDER BY Item_Price DESC;

OR

(B) Sumit has created a database for a Library. The database includes a table named Library whose column (attribute) names are mentioned below:

B_Id: Shows the unique code for each book.
B_Name: Specifies the book title
B_Author: Specifies the author of the book.
B_Qty: Indicates the quantity of books available in the Library.
B_Date: Specifies the date of book published.

Table: Library

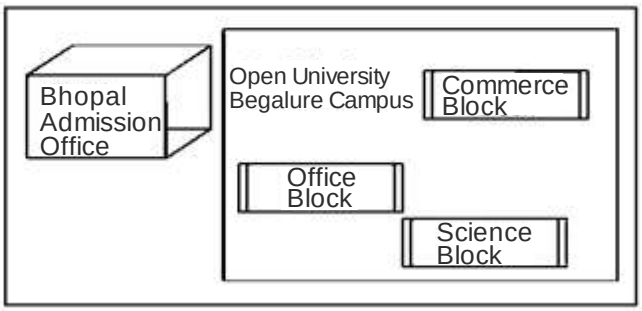
B_Id	B_Name	B_Author	B_Qty	B_Date
B01	The Great Gatsby	F. Scott Fitzgerald	50	1925-10-04
B02	To Kill a Mockingbird	Harper Lee	33	1960-11-07
B03	Pride and Prejudice	Jane Austen	60	1813-01-28
B04	The Catcher in the Rye	J.D. Salinger	20	1951-07-16
B05	Moby Dick	Herman Melville	10	1851-10-18

I Write SQL query for display the number of characters in book name.

II Write SQL query for display the books published in the month of October

	III Write SQL query for displaying the book name with quantity, having the quantity available between 30 to 50 IV Write SQL query for display the name of latest books published.	

	Section-E (2 x 5 = 10 Marks)	
35	Open University of India is starting its first campus in Bengaluru with its centre, admission office in Bhopal. The university has three major blocks comprising Office block, Science block and Commerce block is in a 5 km area campus.	5



As a network expert, you need to suggest the network plan as per (a) to (e) to the authorities keeping in mind the distance and other given parameters.

Expected wire distance between various locations

Blocks	distance
Office Block to Science Block	90 m
Office Block to Commerce Block	80 m
Science Block to Commerce Block	15 m
Bhopal Admission Office to Bengaluru Campus	1400 km

Expected number of computers to be installed at various locations in the university are as follows:

Blocks	Number of Computers
Office Block	10
Science Block	140
Commerce Block	30
Bhopal Admission Office	8

a. Suggest the authorities, the cable layout amongst various blocks inside university campus for connecting the blocks.

b. Suggest the most suitable place (i.e. block) to house the server for this university with a suitable reason.

c. Suggest an efficient device form the following to be installed in each of the blocks to connect all the computers.
(i) Modem (ii) Switch (iii) Gateway
d. Suggest the most suitable (very high speed) service to provide data connectivity between admission office located in Bhopal and the campus located in Bengaluru form the following options:
● Telephone line
● Fixed Line dial-up connection
● Coaxial cable network
● GSM
● Leased line
● Satellite connection.
e. University is planning to connect its campus in Bhopal, which is more than 1400 km. Which type of network will be formed?

36 Consider the DataFrame df shown below. 5

	PlayerID	P_Name	Year	Avg
0	1	Sachin	2010	55.0
1	2	Saurabh	2009	45.0
2	3	Rahul	2011	46.0
3	4	Virat	2020	50.0
4	5	Kapil	1990	40.0

Write Python statements for the DataFrame df to :
I. Remove the column Year.
II. Rename the column name 'P_Name' to 'Player'.
III. Print the first three rows of the DataFrame df.
IV. Display the data of the ‘P_Name' column from indexes 1 to 3 (both included)
V. Display P_Name of all the players.

37 (A) Write suitable SQL query for the following: 5
i. Display 7 characters extracted from 7th left character onwards from the string ‘INDIA SHINING’.
ii. Display the position of occurrence of string ‘COME’ in the string ‘WELCOME WORLD’.
iii. Round off the value 78.779 to 2nd decimal place.
iv. Display the remainder of 149 divided by 6.
v. Remove all the expected leading and trailing spaces from a column useridof the table ‘USERS’ and convert to lowercase.

	OR (B) Write suitable SQL query for the following: (i) Display the string "Intel" from the string “Artificial Intelligence”. (ii) Trim the blanks from both ends from the string " madam " . (iii) To display the name of the current month e.g.- December or January from the current date. (iv) To convert "lower" into uppercase letters. (v) To Find the first occurrence of ‘O’ in "HELLO WORLD"	